Theme Detection - updates II

May 2026

Today: from topics to *themes* — a working definition, the lifecycle, and how to track them through time.

Last time: what is a *topic*.

What is a theme? A working definition (1/5)

Theme $=$ **transient**, **cross-sectoral exposure** created by a shared narrative, structural shift, or value-chain relationship.

Transient

Sectors are stable for decades; themes have a *lifecycle* — they emerge, decay, and mutate.

Cross-sectoral

Themes *cut across* sector boundaries, connecting firms conventional tools never group together.

Exposure

What matters is the cash-flow link, not vocabulary: a firm that *mentions* AI without earning AI revenue is marketing, not a thematic investment.

The Vistra case (2024). A US power utility's stock more than tripled — without an AI pivot. Hyperscalers needed electricity; Vistra owned the assets. The standard map had a *category* (utility); the market had a *narrative*.

$\Rightarrow$ *Markets do not care what a company primarily sells; they care about what its future cash flows are exposed to.*

Theme lifecycle: clean energy 2018–2024 (2/5)

Weak signal 2018 → Emerging 2019 → Validated 2020 → Crowded early 2021 → Institutionalized 2021 → Decaying 2022 → Recycled 2023–24

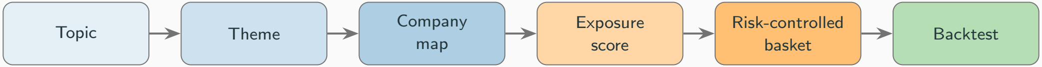
Where alpha lives

Earliest stages — **weak signal** and **emerging** — when a story is detectable but not priced. By the time a thematic ETF reaches the market, the founding observation has often *already been priced*.

What changes per stage

Granularity, breadth of coverage, and the *type of signal* that moves the price: research → flows → index inclusion → liquidation.

From topic to portfolio: the five transitions (3/5)



Each arrow is a research bet.

- Topic → *Theme*: meaning, durability, investability.
- Theme → *Map*: which firms? value-chain reasoning, not vocabulary match.
- Map → *Exposure*: graded score, not membership.
- Exposure → *Basket*: risk targeting, sector neutralisation.
- Basket → *Backtest*: leakage-free, point-in-time.

Why the chain matters

A **cluster of news is not a portfolio**; **a topic is not alpha**. Most apparent alpha in naive setups is the model trading on documents the live system *could not have seen* at decision time.

Tracking themes over time (4/5)

Static models find clusters *after the fact*. **Dynamic** models show that topic vocabulary moves. An **online / candidate-theme** system answers the harder question: is today's cluster the *continuation* of yesterday's, a *split*, a *merger*, or genuinely *new*?

Identity vs. vocabulary drift

Topic *identity* stays fixed; *vocabulary* moves. D-ETM-style models share embedding structure so semantically related words drift together — rather than every rare word having to earn its own history.

Online tracking (BERTrend-style)

Refit/extend the global map only with documents *known as of the decision date*. Keep an **outlier buffer** for proto-themes that have not yet stabilised into a cluster.

Four transitions to detect

Continuation	same theme, drifted vocabulary
Split	one theme branches into several
Merge	several threads converge
Birth	genuinely new candidate theme

False positives to guard against

Duplicate documents, a new source added to the corpus, ticker / OCR errors, seasonal language, coordinated manipulation, model retraining itself.

Better embeddings for the sector-aware pipeline (5/5)

Today (`0.3-themes.ipynb`). FinLang embeds news headlines and 173 ICB subsector definitions → averaged into **45 sector centroids** → BERTopic on headline embeddings → cosine vs centroids → *aligned* (maxcos $\geq \tau_{\text{cov}} = 0.30$) / *novel*. Last run: 22 aligned, 18 novel. *Encoder choice is the single biggest lever — a domain-tuned swap typically beats any pooling or preprocessing tweak.*

Encoder candidates (biggest lever)

- **FinLang** — current baseline (finance-tuned).
- **FinBERT-MPNet** — head-to-head with FinLang on the same 45-sector cosine task.
- **LoRA on `bge-large`** — strong general encoder lightly adapted to financial text; cheaper to maintain than full domain pretraining.

Pooling & preprocessing

- **Sum-pool / concat last 4 hidden layers** of the SBERT encoder, instead of default last-layer mean pooling — cited +10–20% *relative* on topic diversity & coherence on news corpora.
- **Stop-word removal at embedding time**, not only at the c-TF-IDF stage — small alone, compounds with the rest.

Evaluation on the current pipeline. Topic coherence (NPMI, C_v) and diversity on BERTopic output; sector separation on the 45-centroid axis — median `max_cos` for *aligned* vs *novel*, and the share of themes crossing τ_{cov}.